

CAPSTONE PROJECT REPORT

Report 5 – Software Test Documentation

– FPT University, October 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
27/10/2026	M	Lê Nguyễn Quang Đức	Populate concrete test types, tool versions, test team responsibilities, and staging environment details; align test case counts with 82 committed cases in Report5_TestReport.xlsx.

*A - Added M - Modified D - Deleted

II. Testing Documentation

1. Scope of Testing

Test Scope: authorization, learning administration, assignments, program versioning, schematic experiment and symbolic evaluation, feedback/reflection. Physical robot end-to-end and staging deployment are pending validation; no fabricated outcomes.

Test Levels: unit (Jest/Vitest/Python), integration (NestJS service and Prisma/migrations), system/acceptance (four workflow main and exception cases).

Assumptions: seeded isolated database, controlled role-based accounts, no real secrets in repository; robot runtime safety tests must not activate motion outside authorized lab.

2. Test Strategy

Use unit tests for guards/service decisions, integration tests with scoped seeded PostgreSQL fixtures, system tests for UI/API end-to-end flows, and acceptance demonstrations with owned scenarios. Physical robot test results require controlled hardware evidence and safety approval.

Tests cover main and exception paths plus negative cross-organization access, immutable version ownership, race/conflict and dependency unavailability.

2.1 Testing Types

Each test specifies feature/FR, actor, preconditions, steps, expected persisted state, negative result and automation target.

2.2 Test Levels

Unit checks, integration endpoints/transactions, manual staging e2e and fault/retry tests.

Type of Tests	Test Level			
	Unit	Integration	System	Acceptance
Functional & Business Logic	X	X	X	X
Security & Scope Authorization	X	X	X	X
Negative & Boundary Testing	X	X	X	
State Transition & Invariant Verification	X	X	X	

2.3 Supporting Tools

Supporting tools: GitHub Actions, Jest, Vitest, Python unittest/pytest where actually configured, Prisma validation, database fixtures.

Purpose	Tool	Vendor/In-house	Version
Backend unit and integration testing	Jest	Open source	30.5.2
Frontend unit and component testing	Vitest	Open source	2.1.9
Database schema migration & client	Prisma ORM	Open source	5.21.0

3. Test Plan

3.1 Human Resources

Human Resources: Đức test ownership/integration; Đạt learning; Tuấn robot safety/runtime; Lợi assessment/data; independently collect logs.

Worker/Doer	Role	Specific Responsibilities/Comments
Lê Nguyễn Quang Đức	Backend & QA Lead	Auth/RBAC, school administration, robot allocation, and CI test automation
Trần Tiến Đạt	WF-02 Tester	Course delivery, classroom assignment publication, and program submission tests
Vũ Anh Tuấn	WF-03 Tester	Schematic experiment execution, robot runtime client, and safety boundary tests
Phạm Viết Lợi	WF-04 Tester	Assessment, rubric evaluation, student reflection, and AI metadata tests

3.2 Test Environment

Test environments: local React/Vite + NestJS + PostgreSQL/Supabase staging, optional isolated Python runtime. Do not use live production data for destructive tests.

Purpose	Tool	Provider	Version
Frontend Web Application	React 18.3.1 / Vite 5.2.0	In-house	Node.js 22.x
Backend Education API	NestJS 11.0.0 / TypeScri	In-house	Node.js 22.x

	pt 5.9.2		
Relational Database	PostgreSQL 15+ / Supabase	Supabase / Docker	PostgreSQL 15
Robot Device Runtime	Python 3.10+ / FastAPI / LeRobot	In-house	Python 3.10

3.3 Test Milestones

Milestones: Iteration 1 CI/test demonstration before review-2 tag; Defense 1.1 full staging workflows according to BM04-approved targets.

Milestone Task	Start Date	End Date
Iteration 1 QA	Before review-2 tag	Before review-2 tag
Defense 1.1 workflows	After Review 2	Per BM04 approval

4. Test Cases

WF-01 authorized allocation and cross-school denial; WF-02 classroom publishing and student submission; WF-03 schematic pending creation / invalid mode rejection; WF-04 teacher review and unauthorized assessment denial. Each must have preconditions, expected effects, assertions, and execution proof. Report5 spreadsheet holds detailed cases.

Unit Test Cases: Automated Jest suites (192 tests in services/education-api) and Vitest suites (75 tests in web/robot-edu).

- Unit Test Cases: Automated Jest suites (192 tests in services/education-api) and Vitest suites (75 tests in web/robot-edu).
- Integration, System, and State Transition Test Cases: Fully specified in SEP490_FA26SE183_R2_TestReport.xlsx (82 proposed test cases).

5. Test Reports

No verified executed test result supplied in this session. Preserve status Not Run/Pending and record actual run time, commit SHA, log, tester, defect and outcome only after execution.

Test results cannot be described as passed until test commands, run time, commit, logs and defect IDs are captured.

Diagram consistency test cases — Review 2

• ERD-01: Verify all physical PK/FK types and referential constraints from current PostgreSQL migrations.

• ERD-02: Verify optionality and cardinality against nullable and unique constraints.

• STATE-01: Match all four State Machines to implemented enums and service transitions.

• WF-03-01: Education API rejects unsupported execution modes; no physical execution result is assumed.

• WF-01-01: Classroom robot allocation rejects foreign-school and non-entitled resources.